

# NCA Assistant

## User Manual

Version 1.4

Non-Compartmental Pharmacokinetic Analysis

Bioequivalence Testing • Drug Interaction Studies • Power & Sample Size

*Radboud Applied Pharmacometrics*

*Radboudumc, Nijmegen, The Netherlands*

*Designed and developed by Rob ter Heine (hospital pharmacist–clinical pharmacologist,  
Radboudumc)*

Manual version 1.4 — corresponds to NCA Assistant v1.2

## Document Control

| Version | Date       | Author | Changes                                                                                                                                                                                                                                                                                                                                                              |
|---------|------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | March 2026 |        | Initial release                                                                                                                                                                                                                                                                                                                                                      |
| 1.1     | April 2026 |        | Added theoretical background (Part V), expanded statistical methods (Part VI), DDI study analysis, decimal separator guidance                                                                                                                                                                                                                                        |
| 1.2     | April 2026 |        | Quick Start data preparation callout, grid plot fix documentation, single-subject Analysis Record, DDI interpretation thresholds, regulatory DDI guideline links, troubleshooting updates                                                                                                                                                                            |
| 1.3     | April 2026 |        | New chapter 3 (Visualize Data); chapters 3–5 renumbered to 4–6; hub layout updated to 3+3 grid; attribution updated                                                                                                                                                                                                                                                  |
| 1.4     | April 2026 |        | Factual corrections (Rule 6 label, AUC extrapolation flag); new feature documentation (steady-state in BE, balance check, design summary, Apply LLOQ button, excluded profiles note); beginner clarity additions (LLOQ/BLQ definitions, extravascular explained, $R^2$ failure consequences, SHA-256 explained, half-life review how-to, AUC % Extrapolated meaning) |

## Table of Contents

|                                                                  |    |
|------------------------------------------------------------------|----|
| Document Control .....                                           | 2  |
| Table of Contents.....                                           | 3  |
| Quick Start Guide.....                                           | 5  |
| Part I: Introduction to Non-Compartmental Analysis.....          | 6  |
| What Is Non-Compartmental Analysis? .....                        | 6  |
| The Concentration–Time Curve .....                               | 6  |
| Key Pharmacokinetic Parameters .....                             | 6  |
| Why NCA Is Used So Widely .....                                  | 6  |
| When to Use NCA vs. Compartmental Modelling .....                | 6  |
| Part II: Preparing Your Data.....                                | 8  |
| General Format Rules .....                                       | 8  |
| Decimal Separators and Regional Formats.....                     | 8  |
| Scenario 1: Single Subject .....                                 | 8  |
| Scenario 2: Multiple Subjects, Same Dose .....                   | 8  |
| Scenario 3: Multiple Subjects, Different Doses .....             | 8  |
| Scenario 4: Standard 2-Period Crossover (BE) .....               | 9  |
| Scenario 5: Fixed-Order, 3-Period, and Replicate Crossover ..... | 9  |
| Scenario 6: Parallel-Group BE .....                              | 9  |
| Scenario 7: Steady-State PK.....                                 | 9  |
| Scenario 8: Drug–Drug Interaction Study.....                     | 9  |
| BLQ Handling .....                                               | 9  |
| Common Mistakes.....                                             | 10 |
| Part III: Using the App.....                                     | 11 |
| The Hub Screen .....                                             | 11 |
| Path 1: Plan a Study.....                                        | 11 |
| Path 2: Upload & Check Data .....                                | 11 |
| Path 3: Visualize Data .....                                     | 17 |
| Path 4: Analyze One Subject at a Time .....                      | 17 |
| Path 5: Analyze All Subjects (Batch NCA) .....                   | 14 |
| Path 6: Bioequivalence Testing.....                              | 14 |
| Analysing Drug–Drug Interaction Studies .....                    | 14 |
| How to Set Up a DDI Analysis .....                               | 14 |
| Interpreting DDI Results .....                                   | 14 |
| Practical Considerations .....                                   | 15 |
| Complete Analysis Record .....                                   | 15 |
| Part IV: Worked Examples.....                                    | 16 |
| Exercise 1: Theoph Dataset (Batch NCA).....                      | 16 |
| Exercise 2: Bioequivalence .....                                 | 16 |

|                                                               |    |
|---------------------------------------------------------------|----|
| Exercise 3: Power Calculation.....                            | 16 |
| Part V: Theoretical Background .....                          | 17 |
| Trapezoidal Methods for AUC Calculation .....                 | 17 |
| The Linear Trapezoidal Rule .....                             | 17 |
| The Logarithmic Trapezoidal Rule.....                         | 17 |
| The Linear-Up / Log-Down Hybrid .....                         | 17 |
| Terminal Phase Characterisation and Half-Life Estimation..... | 17 |
| What Is the Terminal Phase? .....                             | 17 |
| Automated Point Selection .....                               | 17 |
| Manual Point Selection .....                                  | 17 |
| BLQ Handling: Pharmacological Rationale .....                 | 18 |
| Steady-State Considerations.....                              | 18 |
| Part VI: Statistical Methods.....                             | 19 |
| The Two One-Sided Tests (TOST) Procedure.....                 | 19 |
| The ANOVA Model for Crossover Designs .....                   | 19 |
| Fixed-Effects vs. Mixed-Effects Models .....                  | 19 |
| Confidence Interval Construction .....                        | 19 |
| Design-Specific Models.....                                   | 19 |
| Drug–Drug Interaction Analysis.....                           | 19 |
| Power and Sample Size for Bioequivalence .....                | 20 |
| Regulatory Guidelines .....                                   | 20 |
| Appendices .....                                              | 21 |
| Appendix A: Glossary .....                                    | 21 |
| Appendix B: Recognised Column Names .....                     | 21 |
| Appendix C: Troubleshooting .....                             | 21 |
| Appendix D: Software References .....                         | 22 |

Update: right-click → Update Field → Update Entire Table.

## Quick Start Guide

**Before you begin:** make sure your data file is in the right format. If this is your first time, read Part II (Preparing Your Data) before uploading. Incorrect formatting is the most common reason for errors.

**Step 1.** Open the app at `robterheine.shinyapps.io/NCAassistant` or run `source("install_and_run.R")` in RStudio.

**Step 2.** Click **Upload & Check Data**. Upload your CSV or Excel file. The app auto-detects column names.

**Step 3.** If your data contains BLQ values: set the LLOQ value and select a BLQ rule. If the app detects BLQ entries automatically, a blue “Apply LLOQ = X and process” button appears — click it to apply in one step. Otherwise enter the LLOQ manually and click **Process Data**.

**Step 4.** Optionally, navigate to **Visualize Data** to inspect individual profiles and the geometric mean curve before running any analysis. Then go to **Analyze All Subjects**. Set the dose and route. Click **Run Analysis on All Subjects**.

**Step 5.** View results. Download Results (Excel) or Download Results (CSV), or generate a Complete Analysis Record.

For bioequivalence or drug interaction studies: go to Bioequivalence Testing (Path 6) instead. For study planning: go to Plan a Study (no data needed). A Data Preparation Guide with example datasets is available from the home screen and from the top navigation bar.

## Part I: Introduction to Non-Compartmental Analysis

### What Is Non-Compartmental Analysis?

Non-compartmental analysis (NCA) is a method for describing the pharmacokinetics of a drug without assuming a specific structural model for how the drug distributes in the body. Instead of fitting a mathematical model, NCA calculates pharmacokinetic parameters directly from the observed concentration–time data using numerical integration and regression.

NCA is the standard approach for analyzing pharmacokinetic data in drug development. It is recommended by the EMA, FDA, and PMDA for bioequivalence studies and is routinely used in Phase I clinical trials, dose-finding studies, and therapeutic drug monitoring research.

### The Concentration–Time Curve

After a drug is administered, blood samples are collected at specified times and the drug concentration is measured. When plotted against time, the curve shows absorption (concentrations rising), distribution (transition), and elimination (concentrations declining). NCA extracts quantitative parameters from this curve without assuming its shape.

### Key Pharmacokinetic Parameters

| Parameter              | Symbol            | What it tells you                                                              |
|------------------------|-------------------|--------------------------------------------------------------------------------|
| Peak concentration     | C <sub>max</sub>  | The highest concentration observed. Relevant for efficacy and safety.          |
| Time of peak           | T <sub>max</sub>  | When the peak occurs. Reflects the rate of absorption.                         |
| Area under the curve   | AUC               | Total drug exposure over time. The most important measure of overall exposure. |
| Terminal half-life     | t <sub>1/2</sub>  | Time for the concentration to halve during the terminal phase.                 |
| Apparent clearance     | CL/F              | Rate of drug elimination, normalized to bioavailability.                       |
| Volume of distribution | V <sub>z</sub> /F | Apparent volume the drug distributes into during the terminal phase.           |
| Mean residence time    | MRT               | Average time a drug molecule stays in the body.                                |

### Why NCA Is Used So Widely

Because NCA does not assume a particular compartmental structure, the same calculation methods apply regardless of whether the drug follows simple one-compartment kinetics or complex multi-phasic elimination. The results depend only on the observed data and the numerical method used, not on model selection. This makes NCA highly reproducible across software platforms and analysts.

Regulatory agencies accept NCA because the methods are well-defined and transparent. NCA is the required approach for bioequivalence studies, where C<sub>max</sub> (rate) and AUC (extent) are the primary endpoints.

### When to Use NCA vs. Compartmental Modelling

Use NCA when you need standard PK parameters from observed data: first-in-human studies, bioequivalence, drug interaction studies, dose proportionality, food effect studies. Use compartmental modelling when you need to simulate unobserved scenarios, when you have sparse sampling, or when you need to understand mechanisms. The two approaches are complementary.

## Part II: Preparing Your Data

### General Format Rules

The app expects data in **long format**: one row per observation. Each row contains a subject identifier, the sampling time, and the measured concentration. Accepted formats: CSV (.csv), tab-separated (.tsv, .txt), and Excel (.xlsx, .xls). Column names are auto-detected.

### Decimal Separators and Regional Formats

The app expects periods as decimal separators (e.g., 12.5, not 12,5). If your data uses commas as decimal separators (common in continental Europe), convert the file before uploading. In Excel: open the file, check that numeric columns display correctly, then save as CSV (Comma delimited). If your regional settings use commas, change the decimal separator in Excel's options (File → Options → Advanced → uncheck 'Use system separators', set Decimal separator to period) before saving. Exception: BLQ text entries such as '<0.5' are recognised with both period and comma decimal separators — no conversion is needed for these.

Tab-separated files (.tsv) with semicolon column separators and comma decimals can be handled if you first convert the decimals. The safest approach: open the file in a text editor, use Find & Replace to change all semicolons to commas (or tabs), then change all decimal commas to periods. Alternatively, use Excel as an intermediary.

### Scenario 1: Single Subject

| Time | Concentration |
|------|---------------|
| 0    | 0             |
| 0.5  | 4.2           |
| 1    | 18.5          |
| 2    | 12.1          |
| 4    | 6.3           |
| 8    | 2.8           |
| 24   | 0.4           |

### Scenario 2: Multiple Subjects, Same Dose

| Subject | Time | Concentration |
|---------|------|---------------|
| S001    | 0    | 0             |
| S001    | 1    | 22.5          |
| S001    | 24   | 0.4           |
| S002    | 0    | 0             |
| S002    | 1    | 28.1          |
| S002    | 24   | 0.6           |

### Scenario 3: Multiple Subjects, Different Doses



Add a Dose column if subjects received different doses (dose escalation, weight-based dosing).

### Scenario 4: Standard 2-Period Crossover (BE)

Six columns: Subject, Treatment, Period, Sequence, Time, Concentration. Time resets to 0 at the start of each period. Label the Treatment column 'Test' and 'Reference' — the app recognises these labels and uses Reference as the denominator in the ratio. If you use other labels (e.g., 'Drug A' and 'Drug B'), alphabetical order determines which is the denominator. The results table always shows which label was assigned as Test and which as Reference, so verify this matches your intent before interpreting the confidence interval.

| Subject | Treatment | Period | Seq | Time | Conc |
|---------|-----------|--------|-----|------|------|
| S01     | Test      | 1      | TR  | 0    | 0    |
| S01     | Test      | 1      | TR  | 1    | 24.5 |
| S01     | Reference | 2      | TR  | 0    | 0    |
| S01     | Reference | 2      | TR  | 1    | 22.8 |

### Scenario 5: Fixed-Order, 3-Period, and Replicate Crossover

Same six-column format. For fixed-order: Sequence has one value for all subjects. For 3-period: Period goes to 3. For replicate: Period goes to 4, each treatment appears twice.

### Scenario 6: Parallel-Group BE

Four columns: Subject, Treatment, Time, Concentration. No Period or Sequence needed.

### Scenario 7: Steady-State PK

Same as Scenario 2. Pre-dose concentration is above zero. Tick 'Steady-state (drug given repeatedly)' in the app.

### Scenario 8: Drug–Drug Interaction Study

A drug interaction (DDI) study measures the effect of a co-administered drug (the perpetrator) on the pharmacokinetics of the victim drug. The most common design is a crossover where each subject receives the victim drug alone and then with the perpetrator, or a fixed-order design where the victim is given alone first, then with the perpetrator after a washout.

The data format is identical to a crossover BE study (Scenario 4). Use the same six columns. The recommended approach is to label the Treatment column 'Reference' for the victim drug alone and 'Test' for the victim drug with the perpetrator. The app then automatically uses 'Reference' as the denominator and 'Test' as the numerator, producing the correct fold-change ratio. If you use custom labels instead, the app assigns them alphabetically — verify the ratio direction in the results table. See the section on analysing DDI studies in Part III for interpretation guidance.

### BLQ Handling

| Rule   | Description                                                                         |
|--------|-------------------------------------------------------------------------------------|
| Rule 1 | Pre-first-quantifiable = 0, post-last-quantifiable = missing, between = 0 (default) |
| Rule 2 | All BLQ = 0                                                                         |
| Rule 3 | All BLQ = excluded                                                                  |
| Rule 4 | All BLQ = LLOQ/2                                                                    |
| Rule 5 | Before Cmax = 0, after Cmax = missing                                               |
| Rule 6 | Before first quantifiable = LLOQ/2, rest = 0 (for drugs with absorption lag)        |

Every assay has a lower limit of quantification (LLOQ) — the lowest concentration the assay can reliably measure. Samples collected below this threshold are reported as BLQ (Below the Limit of Quantification), meaning the drug was present but the assay could not give a precise number. You must tell the app what the LLOQ value is so it can apply the chosen rule. Set LLOQ > 0 when your data contains BLQ text entries such as “<0.1” or “BLQ”. The app auto-detects the LLOQ from ‘<X’ entries and will offer a one-click “Apply LLOQ = X and process” button — click it to set the value and continue, or enter the LLOQ manually and click Process Data.

## Common Mistakes

**Wide format** — subjects as columns instead of rows. **Time not reset** — in crossover studies, time must restart at 0 per period. **Entering 0 for missing samples** — use a blank cell. **Duplicate time points** — average replicates first. **LLOQ left at 0** — blocks processing when BLQ text is present. **Comma decimals** — convert to period decimals before uploading.

## Part III: Using the App

### The Hub Screen

Six cards for the six workflow paths, arranged in two rows of three. The top row covers planning and data preparation; the bottom row covers the three analysis paths. The top navigation bar contains the Statistical Methods page, Data Guide, and About.

### Path 1: Plan a Study

Calculates sample size or power for bioequivalence studies. Supports ABE, ABEL (EMA), RSABE (FDA), and NTID. Uses the PowerTOST R package.

### Path 2: Upload & Check Data

Upload CSV or Excel, map columns, set BLQ handling, click Process Data. 20+ quality checks. Errors block processing; warnings allow it. New file uploads clear all previous state. After successful processing, a “Your data is ready — what’s next?” panel appears with direct links to Visualize Data, Single Subject, Batch Analysis, and Bioequivalence.

### Path 3: Visualize Data

Creates publication-ready concentration–time plots from uploaded data. Accessible directly from the hub or via the “Visualize Data” button on the Upload page. Requires data to be uploaded and processed first; a prompt is shown otherwise.

Layout: settings sidebar on the left, stacked plot area on the right. There are no tabs — the Plot Type control determines which plot or plots appear.

### Plot Types

The Plot Type radio button has three options. Select “Individual Profiles (spaghetti plot)” to show one concentration–time line per subject, colour-coded by the grouping variable of your choice. This is the standard plot for visualising individual PK variability, checking for outliers, and detecting carry-over in crossover studies. Select “Summary Plot (geometric mean  $\pm$  gCV%)” to show the geometric mean concentration at each time point with error bars representing the geometric standard deviation. When two or more treatment groups are present they are overlaid automatically with distinct colours. Select “Both plots” to render the spaghetti plot followed by the summary plot in a single scrollable view — useful for team reviews where you want to show both data quality and the central tendency in one screen.

Geometric mean is the appropriate measure of central tendency for log-normally distributed PK parameters such as  $C_{\max}$  and AUC. Arithmetic mean  $\pm$  SD is available as an alternative but is labelled with a warning; it may be appropriate for  $T_{\max}$ , which is not log-normally distributed.

### Sidebar Controls

Y-axis scale (Linear or  $\log_{10}$ ): the semi-log view makes the terminal elimination phase linear, which is useful for identifying biphasic profiles or half-life estimation issues. When log scale is active and zero concentrations are present they are omitted from the plot and a note is shown.

Colour lines by (spaghetti plot, dropdown): options are generated from the columns mapped during upload — Subject ID (always available), Treatment, Period, Sequence (each available only when that column was mapped), and None. Use “Sequence” in crossover studies to check for period effects visually.

Normalize to dose — C/Dose (checkbox, conditional): appears only when a Dose column was mapped. When checked, concentration values are divided by the subject’s dose, so the Y axis shows C/Dose (e.g., ng/mL per mg). Useful for overlaying profiles from dose-escalation studies. Subjects with a missing or zero dose are shown as missing. Update the Y-axis label accordingly when this is active, e.g. “C/Dose (ng·mL<sup>-1</sup>·mg<sup>-1</sup>)”.

Colour palette: “Default (app palette)”, “Colorblind-safe (Okabe-Ito)”, or “Grayscale”. For manuscripts destined for print or journals that do not support colour, select Grayscale and export at 600 DPI.

Theme: “Clean (white background)” (recommended for reports), “Minimal”, or “Classic”. Axis labels, plot title, and base font size (8–18 pt) are editable text fields. The default Y-axis label is “Concentration” — edit to add units, e.g. “Cyclosporin (ng/mL)”. The default X-axis label is “Time (h)”.

### Suggested Figure Legend

Immediately above the plot area, the module displays a card titled “Suggested figure legend.” This card contains a ready-to-copy text description of the current figure, automatically generated from your data and settings. It includes the number of subjects and observations, plot type, summary statistic and sample size (for summary plots), error bar definition, treatment group names when a Treatment column is mapped, a count of BLQ observations excluded from the geometric mean, Y-axis scale, and colour grouping for spaghetti plots. The legend updates automatically whenever you change settings. Copy the text, paste it into your report or manuscript, and adapt as needed — add the drug name, time units, and concentration units.

### Exporting a Figure

Export controls are at the bottom of the settings sidebar under the heading “Export Figure” — always visible, no separate tab required. A small “Will export” label above the Download button shows exactly which plot will be downloaded. When “Both plots” is selected, a radio button appears asking which plot to export; only one file is produced per download.

Format: PNG (good for Word documents, presentations, and emails), PDF (vector; text and lines remain sharp at any zoom level; recommended for journal submission when the journal accepts PDF figures), or SVG (scalable vector graphics; open and edit in Inkscape, Adobe Illustrator, or similar). Width and height are set in inches (2–20 in each); typical journal requirements are 3.0–3.5 in for a single column and 6.5–7.0 in for double column.

Resolution: 150 DPI (screen/drafts), 300 DPI (print, default, minimum for most journals), or 600 DPI (high-resolution journal submission). The Download button is grey until the first plot has rendered in the main panel; it turns green once a plot is visible. The exported filename includes the plot type (spaghetti\_plot.png or summary\_plot.png) so you can download both without overwriting either.

### Interactive Preview vs. Static Export

The plots shown in the main panel are interactive (built with plotly): hover over any point or line to see subject ID, time, and concentration; zoom by clicking and dragging; pan with the pan icon; reset with the home icon. These interactive features are not available in the exported static file. The static file (PNG/PDF/SVG) is built from a separate ggplot2 render using your sidebar settings — font size, theme, and colour palette — and is the version suitable for reports and manuscripts.

## Worked Examples

**Example 1: Quick data check after upload.** Upload `example_theoph.csv` from the Data Preparation Guide. Navigate to “3. Visualize Data”. The individual profiles plot is shown by default — 12 coloured lines, one per subject, on a linear Y scale. Toggle Y-axis to  $\text{Log}_{10}$ : the terminal phase becomes linear, confirming mono-exponential disposition. This is a visual check only; no analysis settings are required.

**Example 2: Summary plot for a crossover BE study.** Upload your crossover dataset with a Treatment column mapped. Navigate to “3. Visualize Data” and select “Summary Plot.” The plot automatically overlays two curves, one per treatment, with error bars. Substantially overlapping error bars are consistent with bioequivalence; clearly separated curves suggest the 90% CI may fall outside 80–125%. Go to the Export Figure section in the sidebar, select PDF, 7 × 5 in, 300 DPI, and click Download. Include the PDF directly in your clinical study report.

**Example 3: Dose-escalation spaghetti plot.** Upload data with a Dose column and different doses per subject. Enable “Normalize to dose (show C/Dose)” in the sidebar. Set Colour lines by “Treatment” (or use Dose group if mapped as Treatment). Update the Y-axis label to “C/Dose ( $\text{ng}\cdot\text{mL}^{-1}\cdot\text{mg}^{-1}$ )”. If dose disposition is proportional, the normalised profiles collapse onto each other; divergence suggests non-linear pharmacokinetics.

## Common Questions

**The Download button is grey.** The button is in the Export Figure section of the sidebar and activates once the first plot has rendered in the main panel. There is no separate tab to visit first — simply wait for the plot to appear.

**Some time points are missing from my summary plot.** The summary plot uses the nominal time values present in your data. If subjects have observations at slightly different actual times (e.g., 0.95 h vs 1.05 h for a nominal 1 h sample), round to nominal times before uploading. The app does not perform nominal-time imputation.

**I see “N observations with concentration  $\leq 0$  excluded.”** This is expected when your data contains BLQ values coded as 0 after applying BLQ rules 1, 2, or 5. These zeros cannot be log-transformed and are excluded from the geometric mean. They are not deleted from your data — only excluded from this summary calculation.

**My spaghetti plot has too many colours to distinguish.** Try the Okabe-Ito (colorblind-safe) palette. With more than eight groups, consider switching to Grayscale and annotating the exported figure in a graphics application.

## Path 4: Analyze One Subject at a Time

Step through individual profiles. Manual data entry available. Half-life inspector allows point adjustment. Downloads: CSV and Complete Analysis Record (zip with reproducibility script).

## Path 5: Analyze All Subjects (Batch NCA)

Settings: route, dose (single or per-subject), steady-state, trapezoidal method, minimum  $R^2$ . Result tabs: concentration–time profiles, parameter table (expandable to 37 columns; AUC % Extrapolated is flagged in amber when it exceeds 20%), summary statistics, individual grid (interactive, paginated, 12 subjects per page), half-life review. If any profiles are excluded because they contain fewer than 3 positive concentration values, a persistent amber note appears above the results tabs listing the affected subjects — check the raw data for those subjects. Downloads: CSV, Excel, Complete Analysis Record.

## Path 6: Bioequivalence Testing

Runs NCA, then ANOVA, then 90% CI. Supports 5 designs: 2-period crossover, fixed-order crossover, 3-period crossover, replicate crossover, parallel. Fixed or mixed-effects model. Standard ABE only (80–125%). Label your Treatment column 'Test' and 'Reference' for unambiguous ratio direction. Custom labels work but the app assigns them alphabetically — check the results table. Tick “Steady-state analysis” if subjects were at steady state — the app then uses  $AUC_{\tau}$  (AUC within the dosing interval) as the primary parameter instead of  $AUC_{\infty}$ . After NCA runs, a design summary card appears showing the detected number of treatments, subjects per arm, periods, and sequences — verify this matches your intended design before interpreting the confidence intervals. If any subjects are missing data for one or more treatments, a persistent amber warning appears; resolve any imbalance before proceeding. Downloads: Excel, CSV, Complete Analysis Record.

## Analysing Drug–Drug Interaction Studies

The app’s Bioequivalence Testing path can be used to analyse drug–drug interaction (DDI) studies because the statistical framework is the same: a crossover design comparing two conditions, analysed with an ANOVA model and a 90% confidence interval for the geometric mean ratio.

### How to Set Up a DDI Analysis

Prepare your data as a crossover dataset (Scenario 4 or 5 in Part II). Label the Treatment column 'Reference' for the victim drug given alone and 'Test' for the victim drug given with the perpetrator. This ensures the ratio is calculated as (with perpetrator) ÷ (alone), which directly gives the fold-change in exposure caused by the interaction. Upload the data, map the columns, and select the appropriate crossover design in the Bioequivalence Testing settings.

Run the analysis. The app produces the geometric mean ratio and 90% confidence interval for  $C_{max}$  and AUC. If you labelled the conditions 'Test' and 'Reference', the ratio is Test ÷ Reference: a ratio of 200% means the perpetrator doubled the victim's exposure; a ratio of 50% means it halved it. Always verify in the results table header which label is in the numerator and which is in the denominator.

### Interpreting DDI Results

The interpretation differs from bioequivalence. In a BE study, you want the 90% CI to fall within 80–125%. In a DDI study, you are quantifying the magnitude of the interaction, not testing for equivalence. The regulatory interpretation depends on the drug’s therapeutic index and the clinical context:

- **No clinically significant interaction:** the 90% CI for the AUC ratio falls entirely within 80–125% (the EMA's default threshold for 'no effect').
- **Weak interaction:** the AUC ratio is between 1.25 and 2.0 (or between 0.5 and 0.8 for induction).
- **Moderate interaction:** the AUC ratio is between 2.0 and 5.0 (inhibition) or between 0.2 and 0.5 (induction).
- **Strong interaction:** the AUC ratio exceeds 5.0 (inhibition) or is below 0.2 (induction).

These thresholds are from the FDA and EMA DDI guidance documents. The classification is based on the point estimate of the AUC ratio, not the CI boundaries.

### Practical Considerations

The app reports the ratio as Test ÷ Reference (expressed as a percentage). If you labelled your Treatment column 'Test' and 'Reference' as recommended, the ratio direction is guaranteed: 'Reference' is always the denominator. In a DDI study, label the victim drug alone as 'Reference' and the combination as 'Test'. A reported ratio of 350% then means the perpetrator increased the victim's AUC 3.5-fold. If you used custom labels, check the results table header to confirm which label is in the numerator.

For fixed-order DDI studies (all subjects receive the victim alone first, then with the perpetrator), select 'Fixed-order crossover' as the design. The app uses a paired analysis, which is statistically appropriate but cannot separate period effects from the interaction effect.

The app's forest plot and CI table are directly usable for the DDI results section of a clinical study report or regulatory submission. The 90% CI boundaries are the values reported in DDI summary tables per FDA and EMA guidance.

### Complete Analysis Record

Available in all three analysis paths (single-subject, batch NCA, and BE/DDI). Self-contained zip with: results.xlsx, analysis\_settings.json, reproduce\_analysis.R (standalone script), data\_integrity.txt (SHA-256 hash), analysis\_summary.html, and the original data file. The SHA-256 hash is a unique digital fingerprint of your data file — a 64-character code that changes if even a single character in the file is altered. Regulatory auditors use it to verify that the data file submitted with the report is byte-for-byte identical to the one analysed, providing a tamper-evident link between raw data and results.



## Part IV: Worked Examples

### Exercise 1: Theoph Dataset (Batch NCA)

Download the theophylline dataset from the app: open the **Data Preparation Guide** (from the home screen banner or the top navigation bar) and click **Download theoph.csv** in the Example Datasets section.

Alternatively, if you have R installed: `write.csv(Theoph, "theoph.csv", row.names = FALSE)`

Upload. Set dose = 320, route = Oral / IM / SC (extravascular), method = Linear-up / Log-down. Run. “Extravascular” means the drug was not injected directly into a vein — it had to be absorbed first (oral, intramuscular, or subcutaneous dosing). This affects how clearance and volume of distribution are calculated: because we cannot separate how much drug reached the bloodstream from how fast it was cleared, these parameters are reported as apparent values (CL/F and Vz/F), divided by the unknown oral bioavailability F.

**Questions:** 1) Mean C<sub>max</sub>? 2) Geometric mean AUC to last? 3) Median T<sub>max</sub>? 4) Longest half-life (which subject)? 5) Geometric CV% for AUC<sub>∞</sub>?

Answers: 1) ~8.76 mg/L. 2) ~101.5 h·mg/L. 3) 1.25 h. 4) Subject 6 (~14.3 h). 5) ~22%.

### Exercise 2: Bioequivalence

Download the crossover BE dataset from the **Data Preparation Guide** (click **Download crossover\_be.csv**). Upload. Map columns. Set dose = 100, design = 2-period crossover. Run.

### Exercise 3: Power Calculation

Plan a Study. ABE, standard 2-period crossover, CV = 25 (enter as percentage), GMR = 95 (enter as percentage), target power = 80 (enter as percentage).

**Questions:** 1) Subjects needed? 2) Power with 30 subjects? 3) Subjects needed at CV = 35%?



## Part V: Theoretical Background

### Trapezoidal Methods for AUC Calculation

#### The Linear Trapezoidal Rule

Between consecutive observations at  $t_i$  and  $t_{i+1}$  with concentrations  $C_i$  and  $C_{i+1}$ , the area is approximated as a trapezoid:

$$\text{AUC}(i) = (t(i+1) - t(i)) \times (C(i) + C(i+1)) / 2$$

This assumes linear concentration change. During elimination, drug concentrations typically decline exponentially, and the linear assumption leads to systematic overestimation: a straight line between two points on an exponential decay lies above the true curve.

#### The Logarithmic Trapezoidal Rule

If the concentration declines mono-exponentially between two points, the exact integral yields:

$$\text{AUC}(i) = (t(i+1) - t(i)) \times (C(i) - C(i+1)) / (\ln C(i) - \ln C(i+1))$$

This gives a more accurate estimate during the terminal phase. It requires both concentrations to be positive.

#### The Linear-Up / Log-Down Hybrid

Proposed by Yeh and Kwan (1978): use the linear formula during the ascending phase (no exponential assumption valid) and the log formula during the descending phase (mono-exponential assumption appropriate). This is the default in the NCA Assistant, in Phoenix WinNonlin, and in most regulatory submissions.

### Terminal Phase Characterisation and Half-Life Estimation

#### What Is the Terminal Phase?

After distribution is complete, drug elimination follows  $C(t) = C_0 \cdot e^{-\lambda_z \cdot t}$ . On a log-linear plot, this appears as a straight line with slope  $-\lambda_z$ . The half-life is  $t_{1/2} = \ln(2) / \lambda_z$ .

#### Automated Point Selection

The app considers all combinations of the last  $n$  points after  $C_{\max}$  ( $n \geq 3$ ), fits OLS regression to each, and selects the combination with the highest adjusted  $R^2$ . The  $R^2_{\text{adj}}$  threshold (default: 0.70) controls how strict the selection is. If no combination meets the threshold, the half-life is not estimable and will appear as NA in the parameter table — along with all parameters that depend on it:  $\text{AUC}_\infty$ ,  $\text{CL}/F$ , and  $\text{Vz}/F$ . This can happen when the terminal phase is poorly sampled, when late samples are near the LLOQ and noisy, or when elimination is genuinely multi-phasic. Lowering the  $R^2$  threshold slider will allow a noisier fit; raising it requires a cleaner log-linear decline.

#### Manual Point Selection

Common situations requiring manual adjustment: multi-phasic elimination, flip-flop kinetics, enterohepatic recirculation, late samples near LLOQ. Use the Half-Life Review tab to inspect and adjust. To adjust manually: select a profile from the dropdown, then tick or untick the checkboxes next to individual time points in the terminal phase panel — only points after

$C_{xax}$  are offered. When satisfied with the selection, click “Recalculate”. If the adjusted  $R^2$  of the selected points falls below the threshold set by the slider (default 0.70), the app will not estimate  $\lambda_z$  automatically, and half-life,  $AUC_\infty$ ,  $CL/F$ , and  $V_z/F$  will show as NA for that profile. A wrong  $\lambda_z$  propagates through all of these derived parameters. AUC % Extrapolated is the percentage of  $AUC_\infty$  that is estimated beyond the last measured concentration rather than calculated from observed data — it reflects how much the result depends on the extrapolation assumption. The parameter table flags values above 20% in amber. This threshold is a widely used convention: extrapolation beyond 20% means the estimated terminal phase is driving a substantial portion of the total AUC, which increases uncertainty. When this flag appears, verify the terminal phase selection and consider reporting only AUC to the last measurable concentration ( $AUC_{xax}$ ) as the primary endpoint.

## BLQ Handling: Pharmacological Rationale

Every assay has a lower limit of quantification (LLOQ). Samples below it are reported as BLQ (Below the Limit of Quantification). The handling choice affects AUC: zero inflates early AUC, missing truncates late AUC, LLOQ/2 is a compromise. Rule 1 (default, Beal 2001) is the most widely used. Rule 6 (LLOQ/2 for all BLQ samples before the first quantifiable observation, then zero) is for drugs with an absorption lag, where several early samples may all be BLQ before the drug appears in the bloodstream. Sensitivity analysis (two rules, compare results) is recommended for regulatory submissions.

## Steady-State Considerations

At steady state,  $AUC_\tau$  replaces  $AUC_\infty$ .  $CL/F = \text{Dose}/AUC_\tau$  is valid.  $AUC_\infty$  and  $V_z/F$  from  $AUC_\infty$  are not meaningful during repeated dosing. Half-life estimation from a single dosing interval can be unreliable if the sampling window does not adequately cover the terminal phase.

## Part VI: Statistical Methods

### The Two One-Sided Tests (TOST) Procedure

Conventional hypothesis testing ( $H_0: \mu_T = \mu_R$ ) can only reject equality, not prove similarity. Schuirmann (1987) reformulated the problem as two one-sided tests:

$$H_{01}: \mu(T)/\mu(R) \leq 0.80 \quad (\text{ratio too low})$$

$$H_{02}: \mu(T)/\mu(R) \geq 1.25 \quad (\text{ratio too high})$$

Both rejected at  $\alpha = 0.05 \rightarrow$  ratio lies within  $[0.80, 1.25] \rightarrow$  bioequivalent. This is equivalent to: the 90% CI for the ratio falls entirely within  $[80\%, 125\%]$ . The limits are symmetric on the log scale:  $\ln(0.80) = -0.2231$ ,  $\ln(1.25) = +0.2231$ .

### The ANOVA Model for Crossover Designs

The standard model for a  $2 \times 2 \times 2$  crossover:

$$\ln(Y) = \mu + \text{SEQ} + \text{SUBJ}(\text{SEQ}) + \text{PER} + \text{TRT} + \varepsilon$$

SEQ accounts for baseline differences between sequence groups. SUBJ removes between-subject variability from the residual. PER accounts for changes between visits. TRT is the effect of interest. The treatment coefficient estimates the log-ratio of geometric means.

### Fixed-Effects vs. Mixed-Effects Models

Fixed-effects (lm): all terms including Subject are fixed. Valid for balanced data. Residual df =  $N - 2$  for  $2 \times 2 \times 2$ .

Mixed-effects (nlme::lme): Subject is a random effect. Handles dropouts, separates between- and within-subject variance. Uses the containment method for denominator df. FDA recommends mixed-effects; EMA accepts both.

### Confidence Interval Construction

$$90\% \text{ CI} = \exp(\delta \pm t(0.95, \text{df}) \times \text{SE}(\delta)) \times 100\%$$

where  $\delta$  is the treatment coefficient (log-scale), SE is from the model's variance-covariance matrix, and df are the denominator degrees of freedom. If the entire interval falls within  $[80.00\%, 125.00\%]$ , bioequivalence is concluded.

### Design-Specific Models

**Fixed-order crossover:**  $\ln(Y) = \mu + \text{SUBJ} + \text{TRT} + \varepsilon$ . Equivalent to a paired t-test. df =  $N - 1$ .

**Parallel design:**  $\ln(Y) = \mu + \text{TRT} + \varepsilon$ . df =  $N_T + N_R - 2$ . Higher variability than crossover.

**3-period and replicate:** Same model with Period having 3 or 4 levels. Standard ABE limits applied.

### Drug–Drug Interaction Analysis

DDI studies use the same statistical framework as BE. The ANOVA model, 90% CI, and geometric mean ratio are computed identically. The difference is interpretation: the ratio

quantifies the fold-change in exposure caused by the interacting drug, and the comparison is against DDI classification thresholds (see Part III) rather than BE acceptance limits.

The FDA Guidance for Industry on Drug Interaction Studies (2020) and the EMA Guideline on the Investigation of Drug Interactions (2012) both recommend reporting the geometric mean ratio and its 90% CI for AUC and  $C_{\max}$  of the victim drug, comparing administration alone vs. with the perpetrator.

## Power and Sample Size for Bioequivalence

Exact power via Owen's Q function for standard ABE. Simulation-based (100,000 studies) for ABEL, RSABE, and NTID. Key inputs: within-subject CV, expected GMR, target power, design. Higher CV → more subjects. GMR further from 1.00 → more subjects. Crossover designs have more power than parallel at the same N.

Practical guidance: assume CV = 30% if no data exist. Assume GMR = 95% (enter 95 in the app). Target power 80% (enter 80) for standard studies or 90% (enter 90) for critical studies. Examine the sensitivity plot for robustness.

## Regulatory Guidelines

**EMA Guideline on Bioequivalence** (CPMP/EWP/QWP/1401/98 Rev. 1, 2010):  
<https://www.ema.europa.eu/en/investigation-bioequivalence-scientific-guideline>

**FDA Statistical Approaches to Establishing Bioequivalence** (2001):  
<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/statistical-approaches-establishing-bioequivalence>

**FDA Bioavailability and Bioequivalence Studies** (2003): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-and-bioequivalence-studies-submitted-ndas-or-inds-general-considerations>

**FDA Clinical Drug Interaction Studies** (2020): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-drug-interaction-studies-guidance-industry>

**EMA Guideline on Drug Interactions** (2012): <https://www.ema.europa.eu/en/investigation-drug-interactions-scientific-guideline>

**ICH M3(R2)**: <https://www.ich.org/page/multidisciplinary-guidelines>

## Appendices

### Appendix A: Glossary

| Abbreviation      | Definition                                                   |
|-------------------|--------------------------------------------------------------|
| ABE               | Average Bioequivalence (fixed 80–125% limits)                |
| ABEL              | Average Bioequivalence with Expanding Limits (EMA, for HVDs) |
| ANOVA             | Analysis of Variance                                         |
| AUC               | Area Under the Curve (total drug exposure)                   |
| AUC <sub>τ</sub>  | AUC within the dosing interval (steady state)                |
| BLQ               | Below the Limit of Quantification                            |
| CL/F              | Apparent clearance                                           |
| C <sub>max</sub>  | Maximum observed concentration                               |
| CV%               | Coefficient of Variation                                     |
| DDI               | Drug–Drug Interaction                                        |
| GMR               | Geometric Mean Ratio (Test/Reference)                        |
| HVD               | Highly Variable Drug (CV > 30%)                              |
| LLOQ              | Lower Limit of Quantification                                |
| MRT               | Mean Residence Time                                          |
| NCA               | Non-Compartmental Analysis                                   |
| NTID              | Narrow Therapeutic Index Drug                                |
| RSABE             | Reference-Scaled Average Bioequivalence (FDA, for HVDs)      |
| TOST              | Two One-Sided Tests                                          |
| T <sub>max</sub>  | Time of maximum observed concentration                       |
| t <sub>1/2</sub>  | Terminal elimination half-life                               |
| V <sub>z</sub> /F | Apparent volume of distribution (terminal phase)             |
| λ <sub>z</sub>    | Terminal elimination rate constant                           |

### Appendix B: Recognised Column Names

| Meaning       | Recognised names (case-insensitive)                  |
|---------------|------------------------------------------------------|
| Subject       | Subject, SubjectID, USUBJID, ID, Pat, PatID, Proband |
| Time          | Time, Hours, Hour, Tpt, NTIM, APTS, Zeit             |
| Concentration | Conc, Concentration, DV, Cp, Result, Konz, Plasma    |
| Treatment     | Treatment, Trt, Formulation, Drug, Arm               |
| Period        | Period, Per, Prd, Phase                              |
| Sequence      | Sequence, Seq, Grp                                   |
| Dose          | Dose, Amt, Amount                                    |

### Appendix C: Troubleshooting

**“Disconnected from the server”** — Analysis exceeded memory or time limits. Run locally in RStudio.

**“Error: An error has occurred”** — Usually NA values reaching a log-scale plot. Check BLQ processing (LLOQ > 0).

**Individual profiles tab is blank** — Fixed in v1.0. If the plot does not appear, reload the page.

**Half-life differs from WinNonlin** — Check  $R^2$  threshold and point selection in the Half-Life Review tab.

**Comma decimals not recognised** — Numeric concentration and time columns require period decimals. Use the CSV import settings (Decimal point selector) or convert before uploading. BLQ text entries (e.g., '<0.5') accept both period and comma decimals automatically.

Large sampling gap warning — For multi-day studies (observation window > 48 h) the threshold is 48 h, so daily trough samples are not flagged. For shorter studies the threshold is 24 h. If the gap shown is close to 24 h, it is normal sampling time variability.

## Appendix D: References

Beal SL. Ways to fit a PK model with some data below the quantification limit. *J Pharmacokinet Pharmacodyn*. 2001;28(5):481–504.

Kim H, et al. NonCompart: Noncompartmental Analysis for Pharmacokinetic Data. *Transl Clin Pharmacol*. 2018;26(1):10–15.

Labes D, Schütz H, Lang B. PowerTOST: Power and Sample Size for (Bio)Equivalence Studies. R package.

Owen DB. A special case of a bivariate non-central t-distribution. *Biometrika*. 1965;52(3/4):437–446.

Pinheiro JC, Bates DM. *Mixed-Effects Models in S and S-PLUS*. Springer, 2000.

Schuirmann DJ. A comparison of the two one-sided tests procedure and the power approach. *J Pharmacokinet Biopharm*. 1987;15(6):657–680.

Yeh KC, Kwan KC. A comparison of numerical integrating algorithms by trapezoidal, Lagrange, and spline approximation. *J Pharmacokinet Biopharm*. 1978;6(1):79–98.

R Core Team. R: A Language and Environment for Statistical Computing. Vienna, Austria.